Jonathan M. Gant

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RESEARCH INTERESTS

Exploring sensory systems and behavior through the lens of natural scene statistics, efficient coding, information theory, and population coding.

EDUCATION

Ludwig Maximilian University of Munich October 2022 – Present
Graduate School of Systemic Neurosciences (GSN) Fast-track Ph.D. in Neuroscience
GPA: 1.14/5.0, where 1.0 is the highest grade

Neuromatch Academy July 2022
Completed the intensive, project-based Computational Neuroscience course

University of Florida August 2018 – May 2022
B.S. in Physics & B.S. in Mathematics
GPA: 3.85/4.0, where 4.0 is the highest grade

Massachusetts Institute of Technology June 2021 – December 2021
MIT Summer Research Program in Biology (MSRP-Bio)
Visiting Student Researcher at the Center for Brains, Minds, and Machines (CBMM)

Universidad Autónoma de Yucatán June 2018 – July 2018
Study abroad in Mérida, México focusing on immersive Spanish learning and tropical ecology studies

HONORS AND AWARDS

COSYNE PRX Life Talk Award	April 2025
COSYNE Presenters Travel Grant	January 2025
University of Florida Sawyer Undergraduate Thesis Award	May 2022
Emerging Scholars Program at the University of Florida	December 2019
Stamps Scholarship at the University of Florida	June 2018

PEER-REVIEWED PUBLICATIONS

1. **Gant, J.M.**, Mlynarski, W.F. "Locomotion optimizes sensory representations through a computational principle shared by rodents and primates." *Science Advances* 12.35 (2026): eaed4172. DOI: 10.1126/sciadv.aed4172

2. **Gant, J.M.**, Banburski A., Deza A. "Evaluating the Adversarial Robustness of a Foveated Texture Transform Module in a CNN." *Shared Visual Representations in Human and Machine Intelligence (SVRHM) Workshop at NeurIPS* (2021). openreview.net/forum?id=HyhSFQ1hOgV.

MANUSCRIPTS IN PREPARATION

1. **Gant, J.M.**, Mlynarski, W.F. "Joint coding of visual and movement signals forms efficient representations for scene inference." (Expected submission: Winter 2026).

RESEARCH EXPERIENCE

Ludwig Maximilian University of Munich June 2023 – Present
Advisor: Professor Wiktor Młynarski *Munich, Germany*

· Investigating joint sensory-behavioral representations for state estimation by extending efficient coding frameworks to account for the modulation of visual inputs by head and eye movements.

· Developed a normative theory based on efficient coding and natural scene statistics to explain modulations of neural coding in the visual systems of diverse species during locomotion.

Technical University of Munich October 2022 – April 2023
Advisor: Professor Julijana Gjorgjieva *Munich, Germany*

· Characterized the effect of inhibitory to excitatory synaptic plasticity mechanisms on the balance of excitation and inhibition in dendritic spiking neural networks.
· Investigated the formation of Hebbian assemblies with different combinations of plasticity mechanisms.

MIT Center for Brains, Minds, and Machines March 2021 – December 2021
Advisors: Dr. Arturo Deza and Professor Tomaso Poggio *Cambridge, MA, USA*

· Implemented a fully differentiable foveated texture transform module for a Convolutional Neural Network (CNN) in PyTorch and showed that the module increased the adversarial robustness of the network for scene classification.
· Investigated the effect stochasticity in biologically inspired mechanisms has on the adversarial robustness and learned representations of CNNs.

University of Florida McKnight Brain Institute December 2019 – April 2022
Advisor: Professor Thomas Mareci *Gainesville, FL, USA*

· Investigated the efficacy of novel edge weight definitions for the analysis of brain structural networks.
· Created the Diffusion Image Network Simulator (DINS), a Python program that simulates diffusion MRI images with complex white matter fiber connectivity patterns.
· Classified and characterized structural brain networks in Alzheimer's disease patients using topological data analysis and machine learning.

CONFERENCES

- **Gant, J.M.**, Młynarski, W. "Joint coding of visual and behavioral signals forms representations of the environment during movement." *Bernstein Conference*. Frankfurt am Main, Germany. September 2026. [Poster]
- **Gant, J.M.**, Młynarski, W. "Efficient representations for vision during active behavior." *COSYNE Conference*. Cascais, Portugal. March 2026. **[Invited Workshop Talk]**
- **Gant, J.M.**, Młynarski, W. "Anticipatory modulation as a unifying principle of sensory coding during locomotion." *Bernstein Conference*. Frankfurt am Main, Germany. October 2025. [Poster]
- **Gant, J.M.**, Młynarski, W. "Anticipatory modulation as a unifying principle of sensory coding during locomotion." *Bernstein Center for Computational Neuroscience Retreat*. Tutzing, Germany. June 2025. [Invited Talk]
- **Gant, J.M.**, Młynarski, W. "Information-preserving modulation as a principle of sensory coding during locomotion across species." *COSYNE Conference*. Montreal, Canada. March 2025. **Watch Talk ▶**. **[Contributed Talk]**
- **Gant, J.M.**, Młynarski, W. "Homeostatic information transmission as a principle for sensory coding during movement." *Bernstein Conference*. Frankfurt am Main, Germany. October 2024. [Poster]
- **Gant, J.M.**, Młynarski, W. "Optimal modulation of sensory neurons during locomotion." *COSYNE Conference*. Lisbon, Portugal. March 2024. [Poster]
- **Gant, J.M.**, DeYoung, D., Mareci, T.H. "Validation of edge weight metrics for brain structural networks using the diffusion image network simulator." *University of Florida College of Medicine Research Day*. Gainesville, FL, USA. April 2022. [Poster]

- **Gant, J.M.**, Banburski A., Deza A. "Evaluating the Adversarial Robustness of a Foveated Texture Transform Module in a CNN." *Shared Visual Representations in Human and Machine Intelligence (SVRHM) Workshop at NeurIPS*. Virtual. December 2021. [Poster]

TEACHING EXPERIENCE

Ludwig Maximilian University of Munich October 2023 – Present
Teaching assistant *Munich, Germany*

· Neural data science
· Psychophysics practical
· Seminar on natural scene statistics and sensory coding

LEADERSHIP AND SERVICE

Talking Science April 2025 – Present
Organizing committee member *Munich, Germany*

· Helped organize the Talking Science event which provides a unique opportunity for graduate students to meet and talk about neuroscience with leading researchers in an informal atmosphere.

Graduate School of Systemic Neurosciences October 2022 – Present
Former student representative *Munich, Germany*

· Helped organize a scientific conference for students by students at the graduate school.
· Organized a scientific retreat for Master's students to the Max Planck Institute of Animal Behavior in Konstanz, Germany.
· Served as a Master's student representative from November 2022 until November 2023.

NeuroCamp August 2024
Academic coordinator *Munich, Germany*

· Helped coordinate the scientific content for a neuroscience summer school for high school students.

University of Florida Chapter of Sigma Pi Sigma October 2020 – April 2022
President, Vice President *Gainesville, FL, USA*

· Helped start the University of Florida chapter of Sigma Pi Sigma, a national physics honor society.

University of Florida Society of Physics Students August 2018 – April 2022
President, Vice President, Treasurer *Gainesville, FL, USA*

· Increased accessibility to science education by providing free tutoring for introductory physics courses and programming tutorials to teach the basics of computer programming in Python.
· Coordinated and participated in an annual demo fair for young children hosted at a local park in recognition of the International Day of Women and Girls in Science.
· Spearheaded a project with Hands On Gainesville and a local high school to design publicly accessible educational materials for young children about physics concepts at play in fun playground activities.

East Gainesville Science Clubs September 2018 – February 2019
Volunteer *Gainesville, FL, USA*

· Taught young children from underrepresented groups about basic scientific concepts and encouraged them to study science.